

Prathamesh Atul Kadam

Pune, Maharashtra | +91 9420026463 | prathameshk2905@gmail.com
prathameshkadam.in | [linkedin.com/in/prathamesh-kadam-pk1229](https://www.linkedin.com/in/prathamesh-kadam-pk1229) | github.com/Pratham-1229

SUMMARY

Computer Engineering student with hands-on Python and C++ projects in backend APIs, CI/CD, Linux, and AI/ML.

EDUCATION

Pune Institute Of Computer Technology

B.E. in Computer Engineering | Overall CGPA: 8.50/10

- Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Pune, Maharashtra

Expected Jul 2027

Dr. D. Y. Patil Junior College

Maharashtra State Board (HSC) 12th Percentage: 85.00%

Pune, Maharashtra

May 2023

C. M. S. English Medium Higher Secondary School

Maharashtra State Board (SSC) 10th Percentage: 88.00%

Pune, Maharashtra

May 2021

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript

Backend & Databases: REST APIs, FastAPI, Node.js, PostgreSQL, MySQL, Redis

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS

Cloud & DevOps: AWS (S3, EC2), Git, GitHub Actions (CI/CD), Linux

AI / ML: LLMs, RAG, Prompt Engineering, OCR, CNNs (DenseNet)

PROJECTS

Gekko – C++ Compiler | C++, x86-64, NASM, Linux, GitHub Actions (CI/CD)

Sep 2026 – Present

- Built a compiler in C++ that translates a custom C-style language into x86-64 assembly, using NASM and `ld` to produce standalone Linux ELF64 executables.
- Implemented a lexer, a precedence-climbing parser producing an AST, and a stack-based code generator supporting variables, nested scopes, arithmetic, and `if/elif/else` control flow.
- Designed a custom arena allocator (contiguous buffer, O(1) pointer-bump allocation, bulk free) for AST nodes, removing per-node `malloc/free` overhead and improving cache locality.
- Built line-numbered diagnostics and scope tracking to detect syntax errors and undeclared identifiers, giving clear debugging feedback.
- Automated testing through CI/CD: 5 unit and integration tests run on every push via GitHub Actions.

BAYMAX – AI Pharmacy Assistant | Python, FastAPI, PostgreSQL, Redis, LangGraph

Feb 2026 – Mar 2026

- Developed Python/FastAPI backend REST APIs with PostgreSQL and Redis and JWT-based authentication, serving patients across web, WhatsApp, SMS, and voice channels.
- Built an LLM-powered agentic engine with LangGraph and Pinecone-based RAG retrieval to automate symptom triage, drug-safety checks, prescription guidance, and follow-up questions; added prescription OCR.
- Implemented monitoring and tracking features, including FDA recall monitoring, medication reminders, adherence tracking, vital logging, admin dashboards, and controlled-drug abuse prevention.

AI-Based Pneumonia Detection System | Python, DenseNet201, CNN

Jan 2025 – May 2025

- Trained and optimized a DenseNet201 CNN in Python for pneumonia detection from chest X-rays, achieving 90% accuracy (AI/ML).
- Integrated the model into a responsive web interface for X-ray upload and real-time prediction, supporting healthcare professionals in low-resource settings.

LEADERSHIP & VOLUNTEERING

Content Head

Aug 2025 – Jul 2026

GameDev Utopia

Pune, Maharashtra

- Grew the club's LinkedIn following by 1,500+ members through targeted content strategy and cross-functional collaboration with design and marketing teams.
- Led and mentored a creative team, maintaining documentation standards, delegating tasks, and running open feedback loops to deliver consistent, high-quality output.

ACHIEVEMENTS

- Finalist at the Mastercard Code for Change Hackathon, among 1,500 students from across Pune.